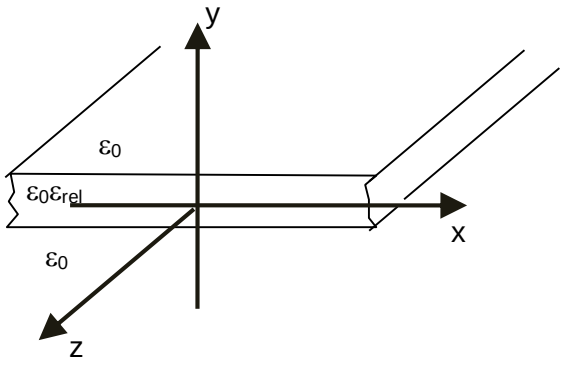
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EXERCISE SESSION 8 (08.01.2021)

Problem 8.1: Dielectric slab modes

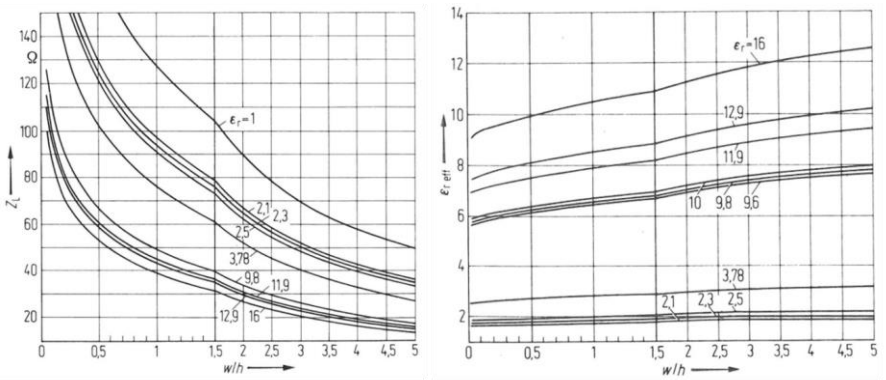
A dielectric slab of infinite extent in the xz-plane and of thickness $2h$ and permittivity ε_{rel} is considered. Assuming guided wave propagation in z-direction and $E_z \neq 0$, $H_z \equiv 0$,

a) name the field components which are not identical to zero
b) determine the functional dependencies of the respective field components in the three domains of the cross-section as functions of the coordinates x,y,z (e.g., "$E_z \sim \exp(-j\beta z) \cdot \sin(k_x x) \cdot \sin(k_y y)$ in the domain y>h" — note that this example is incorrect !)



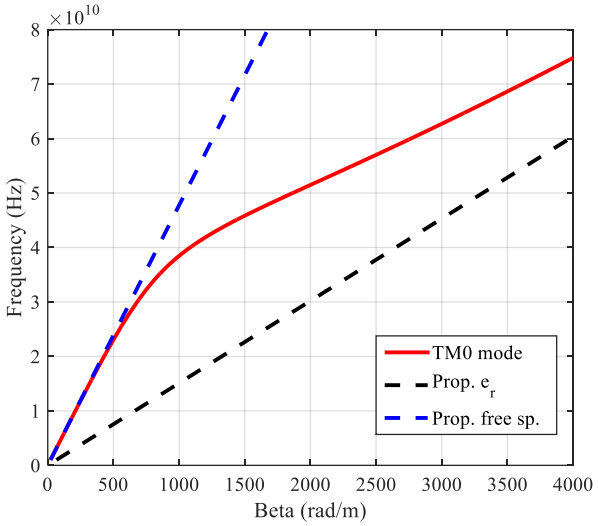
Problem 8.2: Surface wave coupling

A micro-strip thin-film circuit uses alumina ceramic substrate (ε_{rel} = 10, thickness 0.51 mm). Find the frequency where the fundamental slab surface wave mode has the same propagation constant as the micro-strip mode in a 71 Ω line. Graphs for dimensioning of a micro-strip line and disp. graph of the first dielectric slab mode



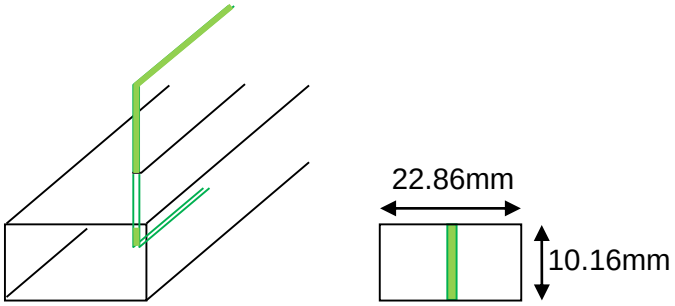
[Zinke Brunswig] , [Wheeler 1977]

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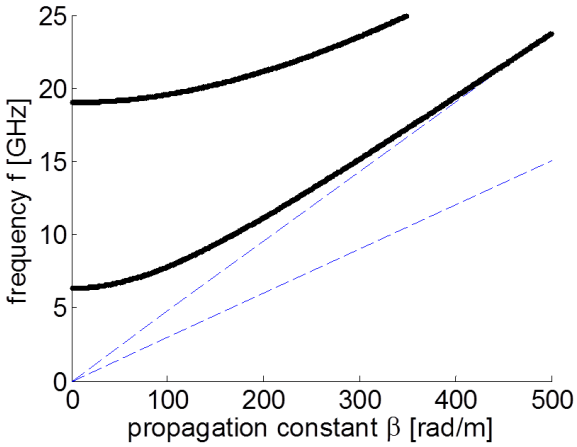
Problem 8.3: Dielectric slab inside a metal waveguide

A mechanically variable phase shifter in rectangular waveguide will be analyzed. A thin dielectric plate is inserted in the waveguide as shown in the picture. The length of the inserted plate is varied mechanically, thereby changing transmission phase.

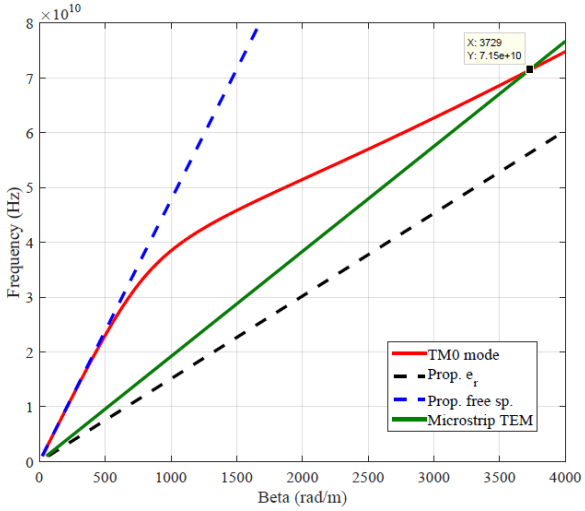


a) Depict qualitatively the distribution of the electric field (both in terms of magnitude and vector direction) in the cross section with the dielectric plate, for a frequency just above cutoff and for a much larger frequency.
b) Assuming a thickness of 0.508mm and a permittivity of 2.5 for the dielectric plate, find the tradeoff between the length of the dielectric inset and the transmission phase for a frequency of 10 GHz (neglecting mismatch at the ends of the dielectric inset). Which ones are the two modes depicted in the graph?

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Solution graph for problem 8.2



Short questions session 8

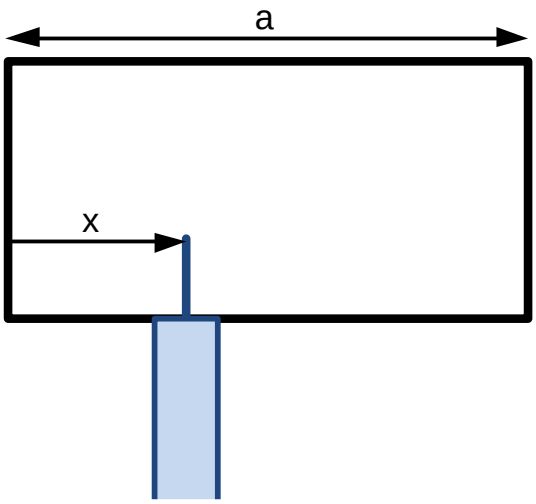
a) A homogenously filled PEC waveguide is considered. Say if the next sentences are right or wrong when considering a certain waveguide mode:

1. H and E field vectors are always perpendicular
2. H field can be normal to the electric conductor boundary
3. E field can be parallel to the electric conductor boundary
4. H field lines form closed loops in the waveguide
5. The fundamental mode is not always a TE

b) Which is the fundamental mode in a grounded dielectric slab? What is its cutoff frequency? From which zero cutoff transmission line technology is coupling to this mode possible?

c) For the waveguide depicted, you can feed it with a coaxial probe in only one of the following points. Propagation is considered normal to the paper.

1. x = 0
2. x = a/6
3. x = a/3
4. x = a/2



d) Where would you place the coaxial probe to feed a rectangular waveguide if you want to excite the TE30 mode without exciting the TE20? Why?

e) Where if you want to excite the TE20 without exciting the TE30?

f) Finally, where would you place it to be able to excite both modes?